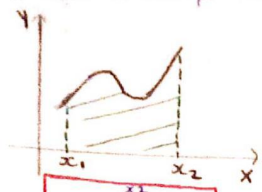
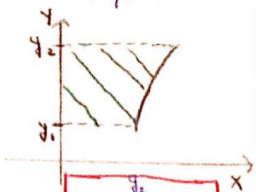
Area Under Curve

1) Area is always +ve and always added.

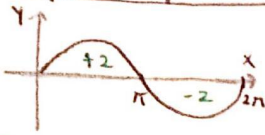


$$\text{Area} = \int_{x_1}^{x_2} y\,dx$$



$$\text{Area} = \int_{y_1}^{y_2} x\,dy$$

2) Definite Integration vis Area ⇒



$$\int_0^{\pi} \sin x\,dx = +2$$

$$\int_{\pi}^{2\pi} \sin x\,dx = -2$$

But, area enclosed with x-axis from $x=0$ to $x=2\pi$ is

$$\int_0^{2\pi} \sin x\,dx = 0$$

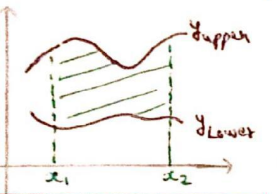
$$\text{Area} = \left|\int_0^{\pi} \sin x\,dx\right| + \left|\int_{\pi}^{2\pi} \sin x\,dx\right| = 2+2 \Rightarrow 4$$

Area Between Two Curves :

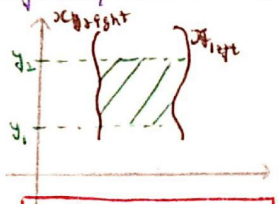
By convention

Area above x-axis / right to y-axis ⇒ +ve

Area below x-axis / left to y-axis ⇒ -ve



$$\text{Area} = \int_{x_1}^{x_2} (y_{upper} - y_{Lower})\,dx$$

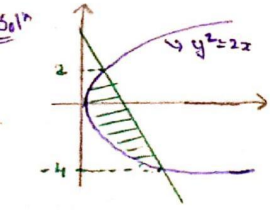


$$\text{Area} = \int_{y_1}^{y_2} (x_{Right} - x_{Left})\,dy$$

(Both concepts are valid in all 4 quadrants)

Ques The area of the region enclosed b/w the parabola $y^2 = 2x$ and the line $x+y=4$ is ____.

Sol^n



$x+y=4 \Rightarrow \boxed{y=4-x}$

$\therefore y^2 = 2x$

(Put) $y^2 = 2(4-y)$

$y^2 + 2y - 8 = 0$

$(y-2)(y+4) = 0$

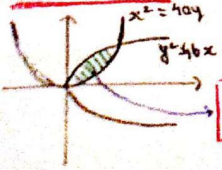
$\Rightarrow \boxed{y = +2, -4}$

$$\text{Area} = \int_{-4}^{2} (x_{Right} - x_{Left})\,dy \Rightarrow \int_{-4}^{2} \left(4 - y - \frac{y^2}{2}\right) dy$$

$$\text{Area} = \left[4y - \frac{y^2}{2} - \frac{y^3}{6}\right]_{-4}^{2} \Rightarrow \left(8 - 2 - \frac{8}{6}\right) - \left(-16 - 8 + \frac{64}{6}\right)$$

$\boxed{\text{Area} = 18}$ Ans

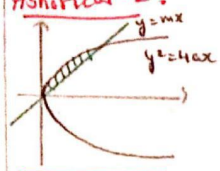
Shortcut -1 :



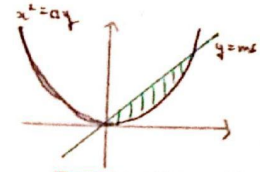
This is applicable only when,
(i) Both parabola have same vertex
(ii) Axis of Parabola ⇒ ⊥ax

$$\text{Area} = \frac{(4a)(4b)}{3} = \frac{16ab}{3}$$

Shortcut -2 :



$$\text{Area} = \frac{8a^2}{3m^3}$$

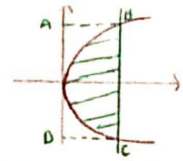


$$\text{Area} = \frac{8a^2m^3}{3}$$

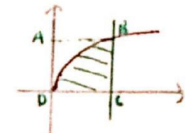
These formulas are applicable only when line also passes through vertex of parabola.

Shortcut -3 :

↳ Area enclosed b/w parabola and double ordinate.



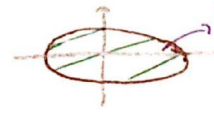
$$\text{Area} = \frac{2}{3}(ABCD)$$



$$\text{Area} = \frac{2}{3}(ABCD)$$

Shortcut -4 :

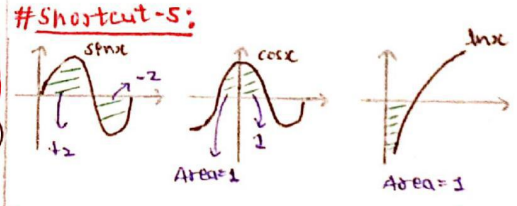
↳ Area enclosed by an ellipse.



$$\text{Area} = \pi ab$$

a, b → semi major/minor axes of ellipse

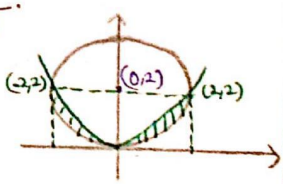
Shortcut -5 :



Ques Area of the region $\{(x,y): x^2 + (y-2)^2 \le 4, x^2 \ge 2y\}$ is ____.

Sol^n $x^2 + (y-2)^2 = 4$ (circle)

$x^2 = 2y$ (parabola)



$x^2 + (y-2)^2 = 4$

$2y + y^2 - 4y + 4 = 4 \Rightarrow y^2 - 2y = 0$

$\boxed{y = 0, 2} \rightarrow x^2 = 2y$, $x^2 = 4$

$\boxed{x = \pm 2}$

Now,

$$\text{Area} = (\text{Area semi circle}) - (\text{Area of parabola})$$

$$= \frac{\pi(2)^2}{2} - \frac{2}{3}(4\times 2)$$

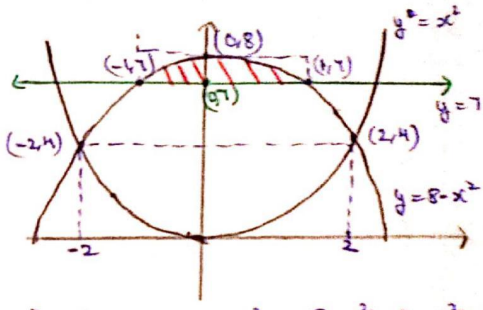
$$\boxed{\text{Area} = 2\pi - \frac{16}{3}}$$ Ans

Ques The area of the region $\{(x,y): x^2 \le y \le 8-x^2, y \le 7y\}$ is ____.

Sol^n $x^2 = y$ (Parabola)

$y = 8 - x^2 \Rightarrow x^2 = 8 - y$ (Parabola)

$y = 7$ (Line)



$\therefore x^2 = y \Rightarrow x^2 = 8 - x^2 \Rightarrow x^2 = 4$

$\boxed{x = \pm 2}$ $\boxed{y = 4}$

$\therefore y = 7 \Rightarrow 8 - x^2 = 7 \Rightarrow \boxed{x = \pm 1}$

Required area = 2 (Area of one parabola) $-\frac{2}{3}$ (Area of Rectangle)

$$\text{Area} = 2 \times \frac{2}{3} \times (4\times 4) - \frac{2}{3}(2\times 1)$$

$\boxed{\text{Area} = 20}$ Ans

Ques The area of the region $\{(x,y): xy \le 8, 1 \le y \le x^2\}$ is ____.

Sol^n $xy = 8$ (Rectangular Hyperbola)

$y = 1$ (Ht. Line)

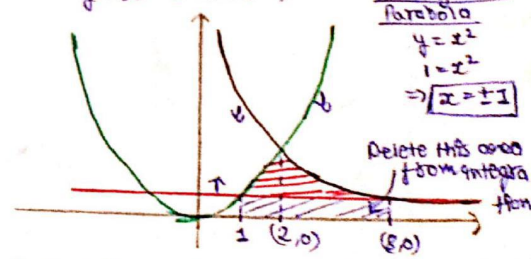
$y = x^2$ (Parabola)

P.O.I of line & parabola

$y = x^2$

$1 = x^2$

$\Rightarrow \boxed{x = \pm 1}$



P.O.I of Parabola & Hyperbola

$xy = 8 \Rightarrow x(x^2) = 8$

$x^3 = 8 \Rightarrow \boxed{x = 2}$

P.O.I of line & Hyperbola

$xy = 8 \Rightarrow x(1) = 8$ i.e. $\boxed{x = 8}$

Now,

$$\text{Req. area} = \int_1^2 x^2\,dx + \int_2^8 \frac{8}{x}\,dx - (7\times 1)$$

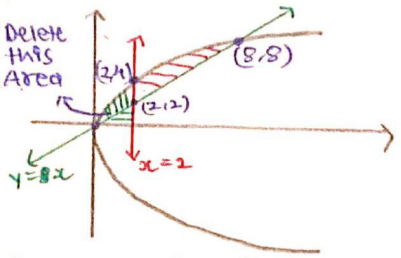
$$= \left[\frac{x^3}{3}\right]_1^2 + 8[\ln x]_2^8 - 7$$

$$= \frac{7}{3} - 7 + 8(2\ln 2)$$

$$\boxed{\text{Area} = 16\ln 2 - \frac{14}{3}}$$ Ans

Ques Let α be the area of longer region bounded by the curve $y^2 = 8x$ and the line $y = x$ and $x = 2$, which lies in the first quadrant. Then the value of 3α is equal to ____.

Soln $y^2 = 8x$ (Parabola) $\quad y = x$ (line) $\quad x = 2$ (line)



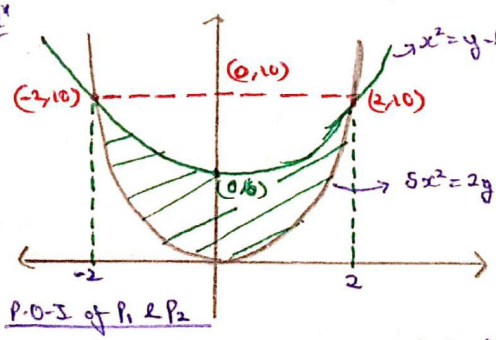
Req. Area $= \frac{8a^3}{3m^2} - \left[\frac{2}{3}(4\times2) - \frac{1}{2}\times2\times2\right]$

$\alpha = \frac{8(2)^2}{3(1)^3} - \left[\frac{16}{3} - 2\right] \Rightarrow \frac{32}{3} - \frac{10}{3}$

$\alpha = \frac{22}{3} \Rightarrow \boxed{3\alpha = 22}$ Ans

Ques If the area enclosed by the parabola $P_1 : 2y = 5x^2$ and $P_2 : x^2 - y + 6 = 0$ is equal to area enclosed by P_1 and $y = \alpha x, \alpha > 0$ then α^3 is equal to ____.

Soln



P.O.I of P_1 & P_2

$5x^2 = 2y \Rightarrow 5x^2 = 2(x^2 + 6)$

$5x^2 = 2x^2 + 12 \Rightarrow 3x^2 = 4$

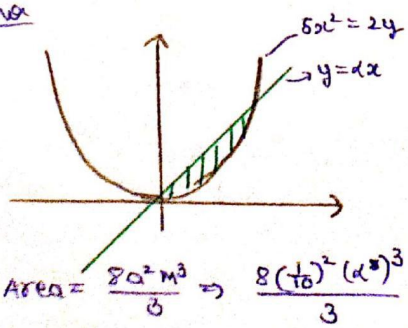
$\Rightarrow \boxed{x = \pm 2}$

Area b/w P_1 & $P_2 = \frac{2}{3}(4\times10) - \frac{2}{3}(4\times4)$

$= \frac{2}{3}(40 - 16) \Rightarrow \frac{48}{3}$

$= \boxed{16}$

Note



Area $= \frac{8a^2m^3}{6} \Rightarrow \frac{8\left(\frac{1}{10}\right)^2(\alpha^3)^3}{3}$

$\frac{8\alpha^3}{300} = 16^2$ (According to ques)

$\boxed{\alpha^3 = 600}$ Ans